LEARNER GUIDE

Deriving a formula

Formulae and Rearranging · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To **derive** a formula is to find it yourself and be able to say **why** it is right. In a growing pattern, look for the part that **stays the same** in every step and the part that **grows** by the same amount each step. The part that stays the same gives the number on its own; the part that grows gives the number × n. Because you can point at each part in the picture, you know the formula works for every step — even step 50.

What you are learning to notice

- A table tells you what the rule is; the picture tells you why.
- The amount added each step is the number in front of n.
- Test your formula on step 1, then on a step you have not drawn.
- Different ways of seeing the same pattern give formulae that look different but always agree, such as $4n + 4$ and $4(n + 1)$.

Moving through it, one step at a time

Draw steps 1, 2 and 3. Colour the part that is the same in all three, and the part that was added. Count the added part: that is the number in front of n. Count the part that stays the same: that is the number on its own. Write the formula, check it against step 1, and use it for a large step.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Squares made from rods

One square uses 4 rods, two squares in a row use 7, three use 10. Derive a formula for the number of rods R in step n.

1. What stays the same? The first rod on the left.	1
2. What grows? Each square adds 3 rods: top, bottom and right.	$3n$
3. Put them together.	$R = 3n + 1$
4. Test on step 50.	$3 \times 50 + 1 = 151$ rods

$R = 3n + 1$, because there is one starting rod and 3 more for each square.

Example 2 — The surface area of a cuboid

A cuboid has edges a, b and c. Derive a formula for its surface area S.

1. The front and back faces are each a × b.	2ab
2. The two sides are each a × c.	2ac
3. The top and bottom are each b × c.	2bc
4. Add the three pairs.	S = 2ab + 2ac + 2bc

S = 2ab + 2ac + 2bc. Each 2 is a pair of matching faces.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (linear sequences and the nth term, and writing formulae from Lesson 3), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.